

Wenlan Gu

Scientific Machine Learning | Optimization | Neural PDE Solvers
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Education

New York University Shanghai

B.S. in Data Science (AI concentration); Minor in Mathematics

Shanghai, China

Expected May 2028

- GPA: 3.815/4.0; study away at NYU New York (Fall 2026 – Spring 2027).
- Coursework: Scientific Computing, Efficient AI Computing (Fall 2026, in progress); Machine Learning, Computer Vision, Linear & Nonlinear Optimization.

Research Experience

Undergraduate Researcher — Optimizer Geometry in Equivariant Force Fields

NYU Shanghai

Prof. Shengjie Wang Lab

Spring 2026 – Present

- **Optimizer inductive bias:** Built MACEBench to study how AdamW, Muon and SOAP shape equivariant force fields; implemented e3nn matrix routing and tensor-product path-aware blocking to test the role of update structure.
- **Loss-landscape geometry:** Implemented force-loss Hessian-vector products, Lanczos/Hutchinson curvature estimates and e3nn block-normalized 2D loss surfaces; built linear mode-connectivity probes with radial-MLP permutation alignment to measure barriers between trained solutions.
- **Spectral mechanisms:** Analyzed weight/update spectra, stable/effective ranks, singular subspaces and representation ranks; used initialization/displacement controls and routed/fallback decomposition to isolate optimizer effects.

DURF Research Fellow — Neural Solvers for Mean-Field Games

NYU Shanghai

Prof. Mathieu Laurière Lab

May 2026 – Present

- **Analytical validation:** Built Riccati/fixed-point and Deep BSDE solvers for stochastic flocking equilibria; used the closed-form linear-quadratic case to validate learned adjoints and warm-start distance-dependent interaction kernels.
- **Convergence analysis:** Investigated sensitivity to training and finite-agent sampling; evaluated terminal BSDE residuals separately from velocity alignment to assess solver accuracy and characterize limits of convergence in nonlinear regimes.

Undergraduate Researcher — Learned Initialization for DFT

NYU Shanghai

Prof. Shengjie Wang Lab

Jan. 2026 – May 2026

- **Scientific solver acceleration:** Contributed to a JAX model predicting constraint-valid initial electron densities for Kohn–Sham DFT; reduced SCF iterations by warm-starting the unchanged solver while retaining converged DFT accuracy.
- **Experimental analysis:** Ran ablations across 13 molecules; built a manifest-driven checkpoint harness yielding a provenance-tracked 60-row benchmark table and distinguished profiling overhead from steady-state GPU cost.

Selected Projects

Inductive Bias of Neural PDE Operators — FNO vs. U-Net

Spring 2026

- Evaluated autoregressive rollouts on PDEBench compressible Navier–Stokes with frequency-band error decomposition; found comparable low-frequency error but up to $301\times$ higher U-Net error in the high-frequency band, exposing failures hidden by aggregate metrics.

Gaussian Mobility Complex — Geometry for Robot Planning

GitHub | 2026

- Developing an SE(2) planner from Gaussian scene/body geometry, orientation-dependent free-space connectivity and continuous path verification; designed robot-conditioned geometric coresets and tested gate preservation against geometry-only compression on synthetic scenes.

Initiator & Co-maintainer — Lab Knowledge Base (MCP + Discord)

2026 – Present

- Built a Python/asyncio + FastMCP service giving researchers and agents access to shared debugging knowledge; implemented SQLite FTS5/BM25 retrieval, validated writes, a serialized Git queue and Discord notifications.

Accepted Workshop Paper

Halluci-action: Evaluating Deception in Claim–Action Alignment of Agents. Co-author.

NeurIPS 2026 BiAlign Workshop — accepted poster. OpenReview

- Co-built a pipeline comparing agents' action claims with tool-execution traces; contributed experiments detecting claimed-but-unperformed and performed-but-undisclosed actions.

Skills & Awards

Programming / ML: Python, SQL, JavaScript, Bash; PyTorch, JAX, scikit-learn, e3nn.

Research computing: Linux, Git, Slurm/HPC, DDP, Singularity, TensorBoard profiling.

Awards: NYU Shanghai Recognition Award (2025, 2026); Dean's Undergraduate Research Fund (2025, 2026).

Languages: Mandarin (native), English (fluent).